

DOCUMENTACIÓN HADOOP

Javier Peralta



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1. Instalamos ssh.

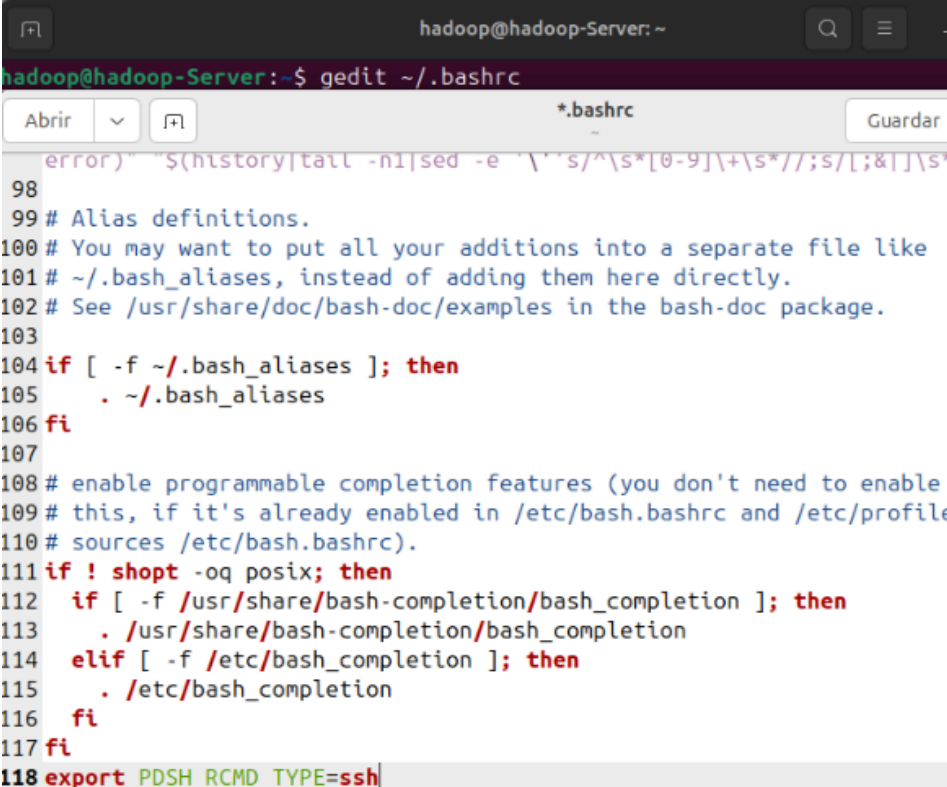
```
hadoop@hadoop-Server: ~  
hadoop@hadoop-Server:~$ sudo apt install ssh  
[sudo] contraseña para hadoop:  
Leyendo lista de paquetes... Hecho  
Creando árbol de dependencias... Hecho  
Leyendo la información de estado... Hecho  
Los paquetes indicados a continuación se instalaron de forma automática y ya no  
son necesarios.  
  libgl1-amber-dri libglapi-mesa libllvm17t64 python3-netifaces  
Utilice «sudo apt autoremove» para eliminarlos.  
Se instalarán los siguientes paquetes adicionales:  
  ncurses-term openssh-server openssh-sftp-server ssh-import-id  
Paquetes sugeridos:  
  molly-guard monkeysphere ssh-askpass  
Se instalarán los siguientes paquetes NUEVOS:  
  ncurses-term openssh-server openssh-sftp-server ssh ssh-import-id  
0 actualizados, 5 nuevos se instalarán, 0 para eliminar y 2 no actualizados.  
Se necesita descargar 837 kB de archivos.  
Se utilizarán 6.801 kB de espacio de disco adicional después de esta operación.  
¿Desea continuar? [S/n]  
Des:1 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-  
server amd64 1:9.6p1-3ubuntu13.14 [37,3 kB]  
Des:2 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-serve  
r amd64 1:9.6p1-3ubuntu13.14 [510 kB]  
Des:3 http://es.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+
```

2. Instalamos pdsh.

```
hadoop@hadoop-Server:~$ sudo apt install pdsh  
Leyendo lista de paquetes... Hecho  
Creando árbol de dependencias... Hecho  
Leyendo la información de estado... Hecho  
Los paquetes indicados a continuación se instalaron de forma automática y ya no  
son necesarios.  
  libgl1-amber-dri libglapi-mesa libllvm17t64 python3-netifaces  
Utilice «sudo apt autoremove» para eliminarlos.  
Se instalarán los siguientes paquetes adicionales:  
  genders libgenders0 ssh-askpass  
Paquetes sugeridos:  
  rdist  
Se instalarán los siguientes paquetes NUEVOS:  
  genders libgenders0 pdsh ssh-askpass  
0 actualizados, 4 nuevos se instalarán, 0 para eliminar y 2 no actualizados.  
Se necesita descargar 201 kB de archivos.  
Se utilizarán 630 kB de espacio de disco adicional después de esta operación.  
¿Desea continuar? [S/n]  
Des:1 http://es.archive.ubuntu.com/ubuntu noble/universe amd64 libgenders0 amd64  
1.22-1build8 [30,8 kB]  
Des:2 http://es.archive.ubuntu.com/ubuntu noble/universe amd64 genders amd64 1.2  
2-1build8 [31,0 kB]  
Des:3 http://es.archive.ubuntu.com/ubuntu noble/universe amd64 ssh-askpass amd64
```

3. Abrimos el archivo .bashrc y añadimos al final del archivo:

```
export PDSH_RCMD_TYPE=ssh
```



```
hadoop@hadoop-Server: ~  
hadoop@hadoop-Server:~$ gedit ~/.bashrc  
Abrir  ▾  *.bashrc  Guardar  
error)" "${history|tail -n1|sed -e '\s/^\s*[0-9]\+\s*//;s/[:;&]]\s'  
98  
99 # Alias definitions.  
100 # You may want to put all your additions into a separate file like  
101 # ~/.bash_aliases, instead of adding them here directly.  
102 # See /usr/share/doc/bash-doc/examples in the bash-doc package.  
103  
104 if [ -f ~/.bash_aliases ]; then  
105     . ~/.bash_aliases  
106 fi  
107  
108 # enable programmable completion features (you don't need to enable  
109 # this, if it's already enabled in /etc/bash.bashrc and /etc/profile  
110 # sources /etc/bash.bashrc).  
111 if ! shopt -oq posix; then  
112     if [ -f /usr/share/bash-completion/bash_completion ]; then  
113         . /usr/share/bash-completion/bash_completion  
114     elif [ -f /etc/bash_completion ]; then  
115         . /etc/bash_completion  
116     fi  
117 fi  
118 export PDSH_RCMD_TYPE=ssh
```

4. Creamos una nueva key para el ssh.

```
hadoop@hadoop-Server:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoop/.ssh/id_rsa):
Your identification has been saved in /home/hadoop/.ssh/id_rsa
Your public key has been saved in /home/hadoop/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:mxN1s1cGJhH9HcpRpgPo0k460r00YkxZjCeNLiesRJg hadoop@hadoop-Server
The key's randomart image is:
+---[RSA 3072]-----+
|.o      ..++o |
|E . o = . .++o |
|. + * =o . +oo.=|
|. . o =. = . =.oo|
|. . o S . . |
|. . o o = . |
|. . +oo= |
|. . .oo... |
|. . ... |
+-----[SHA256]-----+
```

5. Copiamos la nueva key al archivo authorized_keys.

```
hadoop@hadoop-Server:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

6. Verificamos la configuración de SSH.

```
hadoop@hadoop-Server:~$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:RKW+OY1+QSCDeOImdnbStM25Ss9kLLr9UoE82K5Ex5Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

El mantenimiento de seguridad expandido para Applications está desactivado

Se puede aplicar 1 actualización de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
```

7. Instalamos Java 8.

```
hadoop@hadoop-Server:~$ sudo apt install openjdk-8-jdk
[sudo] contraseña para hadoop:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y y
son necesarios.
  libgl1-amd-glx libglapi-mesa libllvm17t64 python3-netifaces
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes adicionales:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11proto-dev
  xorg-sgml-doctools xtrans-dev
Paquetes sugeridos:
  default-jre libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc
  openjdk-8-demo openjdk-8-source visualvm fonts-nanum fonts-ipafont-gothic
  fonts-ipafont-mincho fonts-wqy-microhei fonts-wqy-zenhei fonts-indic
Se instalarán los siguientes paquetes NUEVOS:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-8-jdk
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11proto-dev
```

8. Comprobamos si Java está instalado.

```
hadoop@hadoop-Server:~$ java -version
openjdk version "1.8.0_462"
OpenJDK Runtime Environment (build 1.8.0_462-8u462-ga-us1-0ubuntu2~24.04.2-b08)
OpenJDK 64-Bit Server VM (build 25.462-b08, mixed mode)
```

9. Descargamos Hadoop.

```
hadoop@hadoop-Server:~$ sudo wget -P ~ https://mirrors.sonic.net/apache/hadoop/c
ommon/hadoop-3.4.2/hadoop-3.4.2.tar.gz
--2025-10-30 20:00:56-- https://mirrors.sonic.net/apache/hadoop/common/hadoop-3
.4.2/hadoop-3.4.2.tar.gz
Resolviendo mirrors.sonic.net (mirrors.sonic.net)... 157.131.224.201, 2001:5a8:6
01:4003:157:131:224:201
Conectando con mirrors.sonic.net (mirrors.sonic.net)[157.131.224.201]:443... con
ectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 1065831750 (1016M) [application/x-gzip]
Guardando como: '/home/hadoop/hadoop-3.4.2.tar.gz'

hadoop-3.4.2.tar.gz  0%[                ] 515,54K  142KB/s  eta 2h 5m
```

10. Descomprimos el archivo instalado.

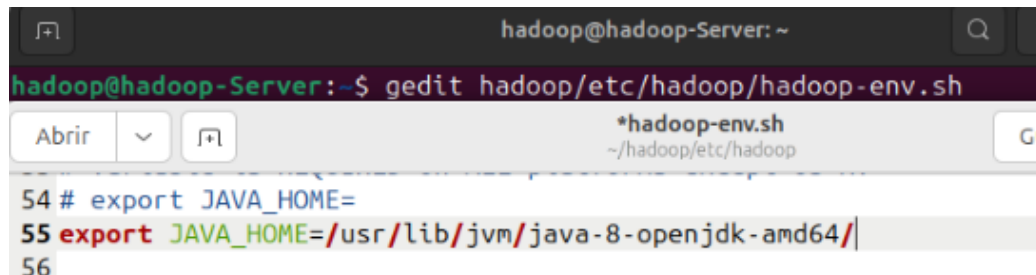
```
hadoop@hadoop-Server:~$ tar xzf hadoop-3.4.2.tar.gz
```


11. Cambiamos el nombre de la carpeta de hadoop-3.4.2 a hadoop.

```
hadoop@hadoop-Server:~$ mv hadoop-3.4.2 hadoop
```

12. Abrimos el archivo hadoop-env.sh y cambiamos la línea export JAVA_HOME por:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/
```



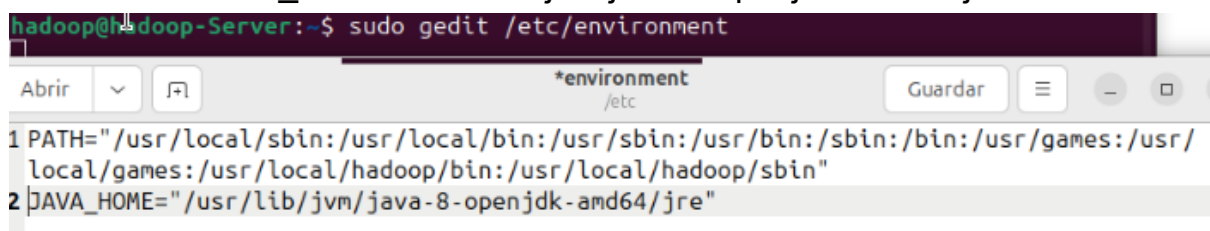
13. Movemos la carpeta hadoop al directorio /usr/local/hadoop

```
hadoop@hadoop-Server:~$ sudo mv hadoop /usr/local/hadoop
```

14. Abrimos el archivo /etc/environment y agregamos las siguientes 2 líneas:

a. PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/local/hadoop/bin:/usr/local/hadoop/sbin"

b. JAVA_HOME="/usr/lib/jvm/java-8-openjdk-amd64/jre"



15. Creamos el usuario hadoopuser.

```
hadoop@hadoop-server:~$ sudo adduser hadoopuser
[sudo] contraseña para hadoop:
info: Añadiendo el usuario 'hadoopuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Añadiendo el nuevo grupo 'hadoopuser' (1001) ...
info: Adding new user 'hadoopuser' (1001) with group 'hadoopuser (1001)' ...
info: Creando el directorio personal '/home/hadoopuser' ...
info: Copiando los ficheros desde '/etc/skel' ...
Nueva contraseña:
CONTRASEÑA INCORRECTA: La contraseña tiene menos de 8 caracteres
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para hadoopuser
Introduzca el nuevo valor, o presione INTRO para el predeterminado
    Nombre completo []: hadoop
    Número de habitación []: 0
    Teléfono del trabajo []: 0
    Teléfono de casa []: 0
    Otro []: 0
¿Es correcta la información? [S/n]
info: Adding new user 'hadoopuser' to supplemental / extra groups 'users' ...
info: Añadiendo al usuario 'hadoopuser' al grupo 'users' ...
```

16. Cambiamos los permisos del usuario hadoopuser.

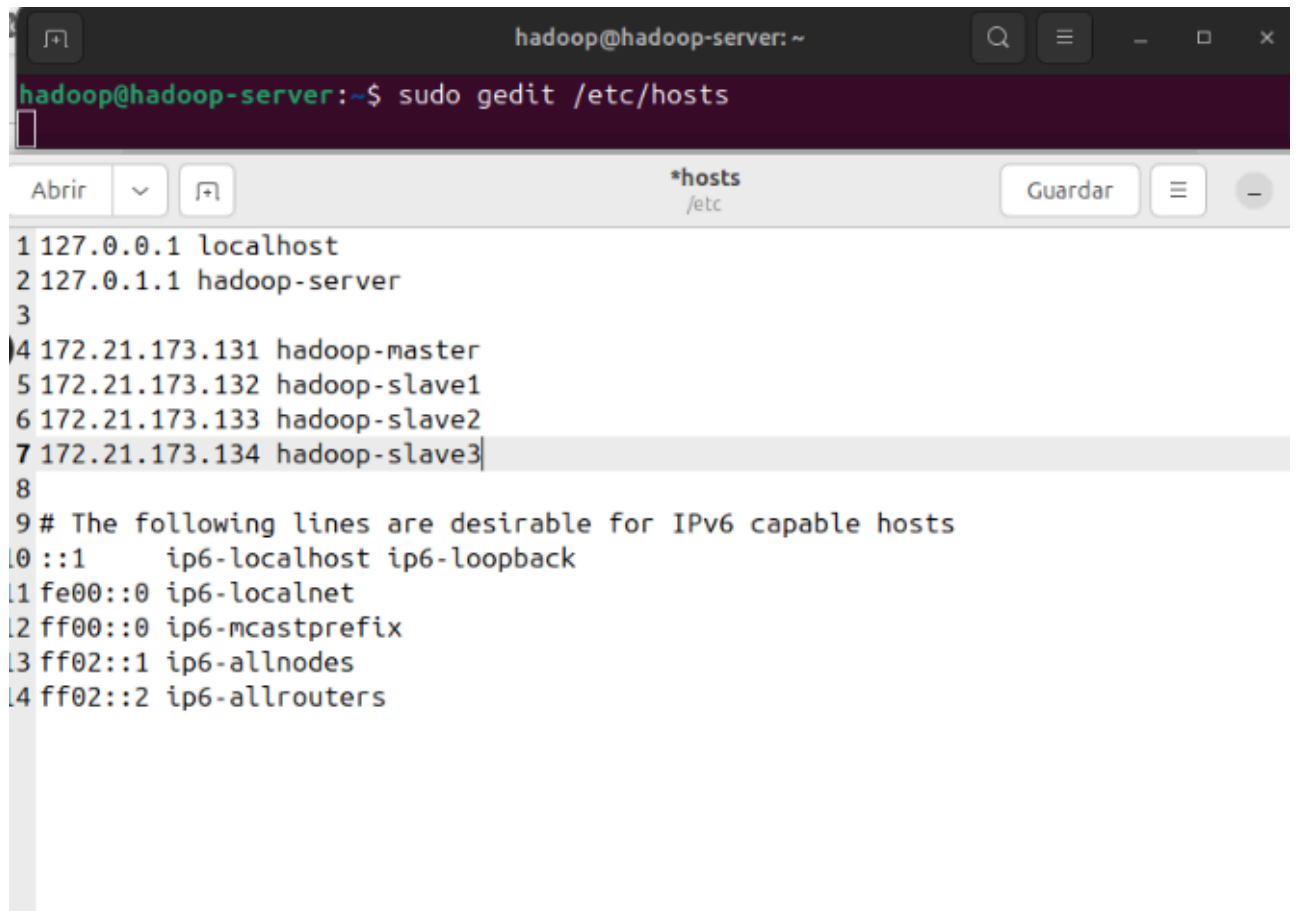
```
hadoop@hadoop-server:~$ sudo usermod -aG hadoopuser hadoopuser
[sudo] contraseña para hadoop:
hadoop@hadoop-server:~$ sudo chown hadoopuser:root -R /usr/local/hadoop/
hadoop@hadoop-server:~$ sudo chmod g+rx -R /usr/local/hadoop/
hadoop@hadoop-server:~$ sudo adduser hadoopuser sudo
info: Añadiendo al usuario 'hadoopuser' al grupo 'sudo' ...
```

17. Verificamos la IP del ordenador.

```
hadoop@hadoop-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:87:c9:b3 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 172.21.173.131/26 brd 172.21.173.191 scope global dynamic noprefixroute ens18
        valid_lft 5634sec preferred_lft 5634sec
    inet6 fe80::be24:11ff:fe87:c9b3/64 scope link
        valid_lft forever preferred_lft forever
```


18. Abrimos el archivo /etc/hosts y agregamos las IP del master y los slaves. En mi caso son:

- a. Master: 172.21.173.131
- b. Slave1: 172.21.173.132
- c. Slave2: 172.21.173.133
- d. Slave3: 172.21.173.134



```
hadoop@hadoop-server: ~  
hadoop@hadoop-server:~$ sudo gedit /etc/hosts  
Abrir  *hosts /etc Guardar  
1 127.0.0.1 localhost  
2 127.0.1.1 hadoop-server  
3  
4 172.21.173.131 hadoop-master  
5 172.21.173.132 hadoop-slave1  
6 172.21.173.133 hadoop-slave2  
7 172.21.173.134 hadoop-slave3  
8  
9 # The following lines are desirable for IPv6 capable hosts  
10 ::1 ip6-localhost ip6-loopback  
11 fe00::0 ip6-localnet  
12 ff00::0 ip6-mcastprefix  
13 ff02::1 ip6-allnodes  
14 ff02::2 ip6-allrouters
```

19. Creamos los slaves clonando la máquina master 3 veces.

Clone VM 21110010 (hadoopServer) 

Nodo de destino:

VM ID:

Nombre:

Conjunto de recursos:

Almacenamiento de destino:

Formato:

 Ayuda

Clonar

Clone VM 21110010 (hadoopServer) 

Nodo de destino:

VM ID:

Nombre:

Conjunto de recursos:

Almacenamiento de destino:

Formato:

 Ayuda

Clonar

Clone VM 21110010 (hadoopServer) 

Nodo de destino:

VM ID:

Nombre:

Conjunto de recursos:

Almacenamiento de destino:

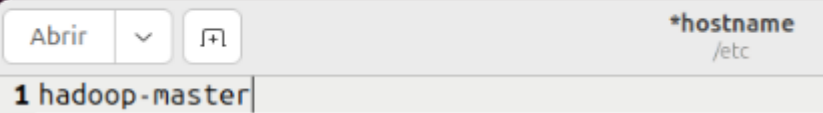
Formato:

 Ayuda

Clonar

20. En el master, abrimos el archivo /etc/hostname y ponemos el nombre de nuestra máquina.

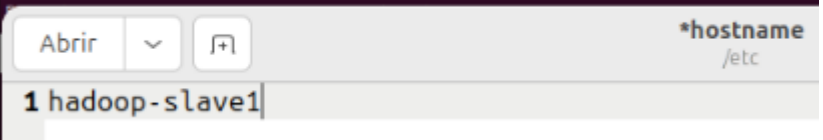
```
hadoop@hadoop-server:~$ sudo gedit /etc/hostname
```



The screenshot shows the gedit text editor window. The title bar indicates the file is `*hostname` located in `/etc`. The editor contains the text `1 hadoop-master` on the first line.

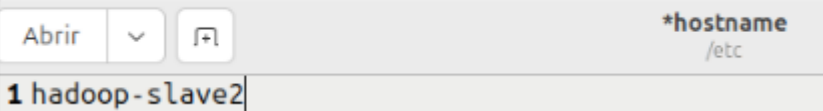
21. Hacemos lo mismo con los slaves.

```
hadoop@hadoop-server:~$ sudo gedit /etc/hostname
```



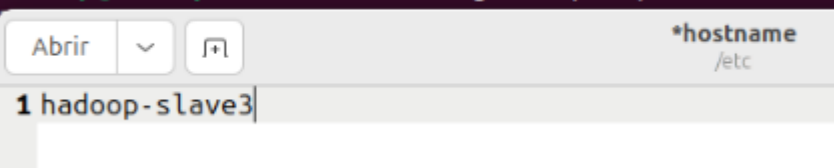
The screenshot shows the gedit text editor window. The title bar indicates the file is `*hostname` located in `/etc`. The editor contains the text `1 hadoop-slave1` on the first line.

```
hadoop@hadoop-server:~$ sudo gedit /etc/hostname
```



The screenshot shows the gedit text editor window. The title bar indicates the file is `*hostname` located in `/etc`. The editor contains the text `1 hadoop-slave2` on the first line.

```
hadoop@hadoop-server:~$ sudo gedit /etc/hostname
```



The screenshot shows the gedit text editor window. The title bar indicates the file is `*hostname` located in `/etc`. The editor contains the text `1 hadoop-slave3` on the first line.

22. Cambiamos de usuario a hadoopuser y creamos una key SSH.

```
hadoopuser@hadoop-master: /home/hadoop
hadoop@hadoop-master:~$ su hadoopuser
Contraseña:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hadoopuser@hadoop-master:/home/hadoop$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoopuser/.ssh/id_rsa):
Created directory '/home/hadoopuser/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hadoopuser/.ssh/id_rsa
Your public key has been saved in /home/hadoopuser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:HpRqFHATmld/nRq/aWMg8qCaEzIglULWx6YYoMkS2wI hadoopuser@hadoop-master
The key's randomart image is:
---[RSA 3072]---+
Eo..o.=..      |
**o. B + o    . . |
*++ * o o . o o |
+o . o o . . +  |
..    o S . o .  |
  o .. o = . . o |
  o      *        |
```

23. Copiamos la key SSH a todos los usuarios.

```
hadoopuser@hadoop-master:~/hadoop$ ssh-copy-id hadoopuser@hadoop-master
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/.ssh/id_rsa.pub"
The authenticity of host 'hadoop-master (172.21.173.131)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kS0/ArJM4+TZcCrDex2dyFFt1Iz+Us0ljaI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-master's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'hadoopuser@hadoop-master'"
and check to make sure that only the key(s) you wanted were added.
```

```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-slave1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/.ssh/id_rsa.pub"
The authenticity of host 'hadoop-slave1 (172.21.173.132)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kS0/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-slave1's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'hadoopuser@hadoop-slave1'"
and check to make sure that only the key(s) you wanted were added.
```

```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-slave2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/.ssh/id_rsa.pub"
The authenticity of host 'hadoop-slave2 (172.21.173.133)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kS0/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-slave2's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'hadoopuser@hadoop-slave2'"
and check to make sure that only the key(s) you wanted were added.
```

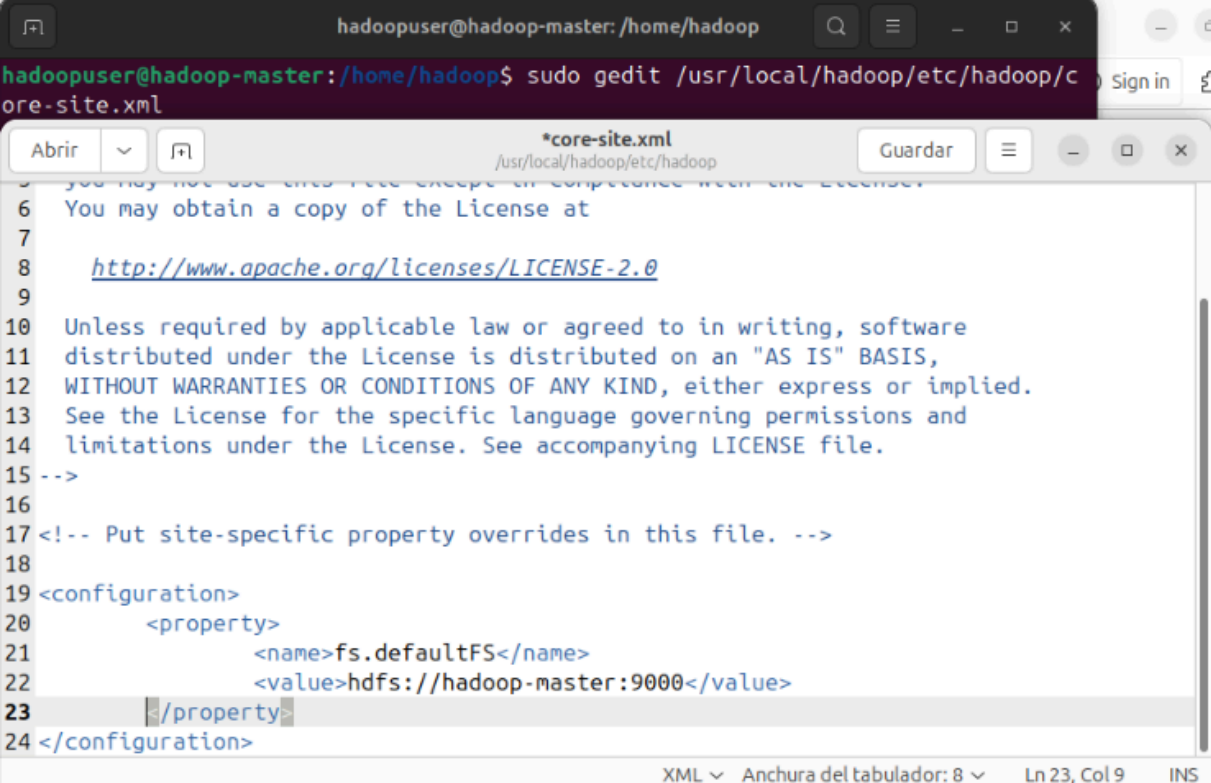
```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-slave3
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/.ssh/id_rsa.pub"
The authenticity of host 'hadoop-slave3 (172.21.173.134)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kS0/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-slave3's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'hadoopuser@hadoop-slave3'"
and check to make sure that only the key(s) you wanted were added.
```

24. En el master, abrimos el archivo core-site.xml y añadimos lo siguiente:

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://hadoop-master:9000</value>
  </property>
</configuration>
```

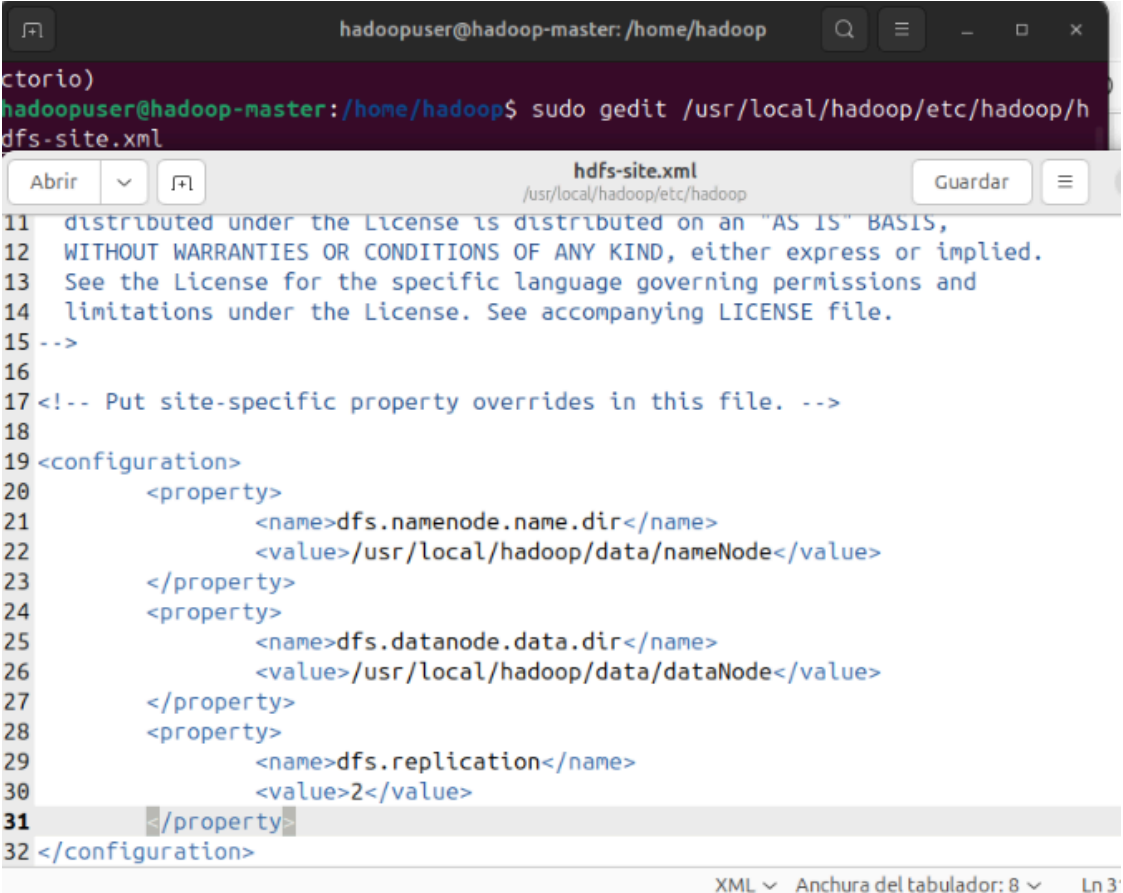


The screenshot shows a terminal window at the top with the command `sudo gedit /usr/local/hadoop/etc/hadoop/core-site.xml` being executed. Below the terminal is a text editor window titled `*core-site.xml` showing the contents of the file. The file contains a license notice and an XML configuration block. The configuration block is being edited, with the cursor at the end of the `</property>` tag on line 23.

```
6 You may not use this file except in compliance with the License.
7 You may obtain a copy of the License at
8 http://www.apache.org/licenses/LICENSE-2.0
9
10 Unless required by applicable law or agreed to in writing, software
11 distributed under the License is distributed on an "AS IS" BASIS,
12 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
13 See the License for the specific language governing permissions and
14 limitations under the License. See accompanying LICENSE file.
15 -->
16
17 <!-- Put site-specific property overrides in this file. -->
18
19 <configuration>
20   <property>
21     <name>fs.defaultFS</name>
22     <value>hdfs://hadoop-master:9000</value>
23   </property>
24 </configuration>
```


25. En el master, abrimos el archivo hdfs-site.xml

```
<configuration>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>/usr/local/hadoop/data/nameNode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>/usr/local/hadoop/data/dataNode</value>
  </property>
  <property>
    <name>dfs.replication</name>
    <value>2</value>
  </property>
</configuration>
```



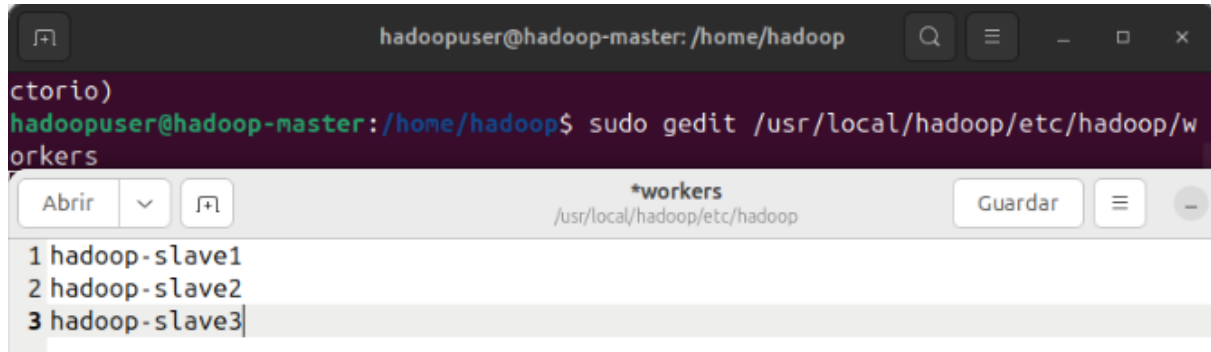
The screenshot shows a terminal window at the top with the command `sudo gedit /usr/local/hadoop/etc/hadoop/hdfs-site.xml` being executed. Below the terminal is a text editor window titled `hdfs-site.xml` showing the XML configuration. The file content includes a license notice and a `<configuration>` block with three properties: `dfs.namenode.name.dir`, `dfs.datanode.data.dir`, and `dfs.replication`. The `dfs.replication` property is currently set to 2. The editor interface includes a toolbar with 'Abrir', 'Guardar', and a search icon, and a status bar at the bottom indicating 'XML', 'Anchura del tabulador: 8', and 'Ln 3'.

```
hadoopuser@hadoop-master: /home/hadoop
torio)
hadoopuser@hadoop-master:/home/hadoop$ sudo gedit /usr/local/hadoop/etc/hadoop/hdfs-site.xml

hdfs-site.xml
/usr/local/hadoop/etc/hadoop

11 distributed under the License is distributed on an "AS IS" BASIS,
12 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
13 See the License for the specific language governing permissions and
14 limitations under the License. See accompanying LICENSE file.
15 -->
16
17 <!-- Put site-specific property overrides in this file. -->
18
19 <configuration>
20   <property>
21     <name>dfs.namenode.name.dir</name>
22     <value>/usr/local/hadoop/data/nameNode</value>
23   </property>
24   <property>
25     <name>dfs.datanode.data.dir</name>
26     <value>/usr/local/hadoop/data/dataNode</value>
27   </property>
28   <property>
29     <name>dfs.replication</name>
30     <value>2</value>
31   </property>
32 </configuration>
```

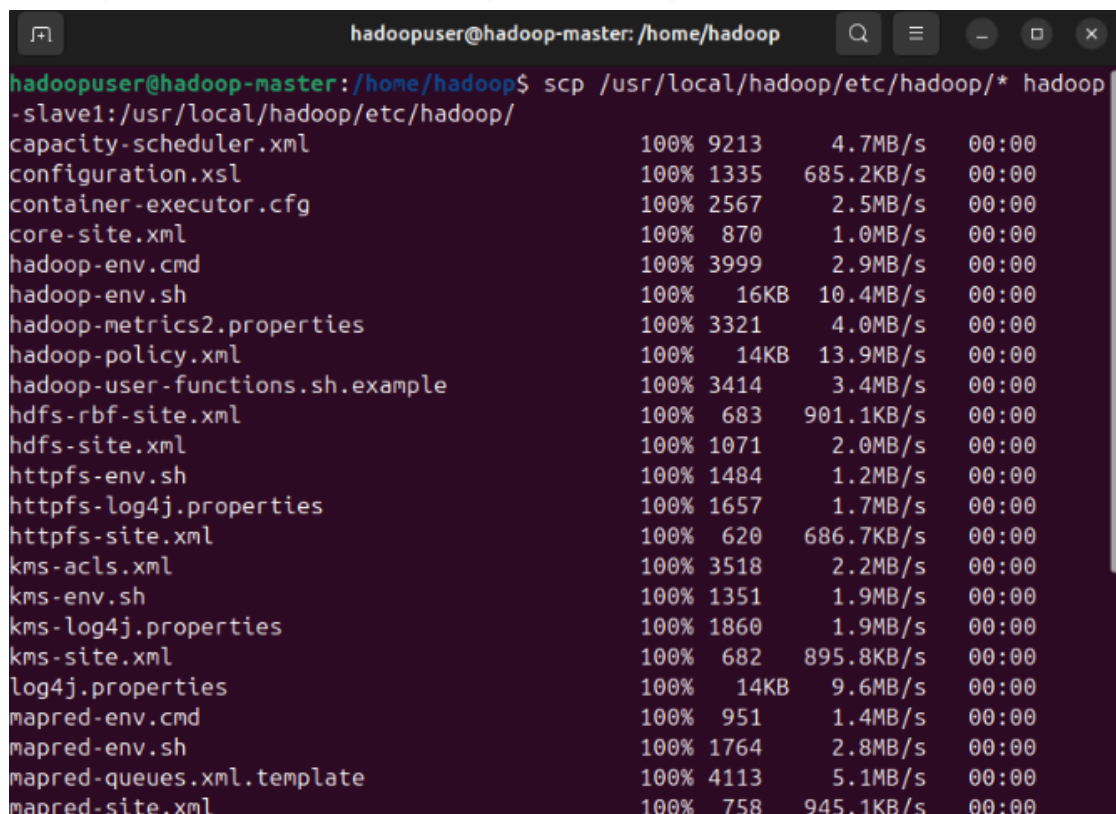
26. Abrimos el archivo workers y ponemos el nombre de los slaves.



```
hadoopuser@hadoop-master: /home/hadoop
ctorio)
hadoopuser@hadoop-master:/home/hadoop$ sudo gedit /usr/local/hadoop/etc/hadoop/workers
Abrir  v  Guardar  -
*workers
/usr/local/hadoop/etc/hadoop/
1 hadoop-slave1
2 hadoop-slave2
3 hadoop-slave3
```

27. Usamos los siguientes comandos para copiar la configuración del master a los slaves (un comando por cada slave):

```
scp /usr/local/hadoop/etc/hadoop/*
hadoop-slave1:/usr/local/hadoop/etc/hadoop/
```



```
hadoopuser@hadoop-master: /home/hadoop
hadoopuser@hadoop-master:/home/hadoop$ scp /usr/local/hadoop/etc/hadoop/* hadoop-slave1:/usr/local/hadoop/etc/hadoop/
capacity-scheduler.xml          100% 9213      4.7MB/s   00:00
configuration.xsl               100% 1335     685.2KB/s  00:00
container-executor.cfg         100% 2567      2.5MB/s   00:00
core-site.xml                   100% 870       1.0MB/s   00:00
hadoop-env.cmd                  100% 3999      2.9MB/s   00:00
hadoop-env.sh                   100% 16KB     10.4MB/s  00:00
hadoop-metrics2.properties      100% 3321      4.0MB/s   00:00
hadoop-policy.xml               100% 14KB     13.9MB/s  00:00
hadoop-user-functions.sh.example 100% 3414      3.4MB/s   00:00
hdfs-rbf-site.xml               100% 683      901.1KB/s 00:00
hdfs-site.xml                   100% 1071      2.0MB/s   00:00
httpfs-env.sh                   100% 1484      1.2MB/s   00:00
httpfs-log4j.properties         100% 1657      1.7MB/s   00:00
httpfs-site.xml                 100% 620      686.7KB/s 00:00
kms-acls.xml                    100% 3518      2.2MB/s   00:00
kms-env.sh                      100% 1351      1.9MB/s   00:00
kms-log4j.properties            100% 1860      1.9MB/s   00:00
kms-site.xml                    100% 682      895.8KB/s 00:00
log4j.properties                100% 14KB      9.6MB/s   00:00
mapred-env.cmd                  100% 951       1.4MB/s   00:00
mapred-env.sh                   100% 1764      2.8MB/s   00:00
mapred-queues.xml.template       100% 4113      5.1MB/s   00:00
mapred-site.xml                 100% 758      945.1KB/s 00:00
```

```
hadoopuser@hadoop-master: /home/hadoop
hadoopuser@hadoop-master: /home/hadoop$ scp /usr/local/hadoop/etc/hadoop/* hadoop
-slave2:/usr/local/hadoop/etc/hadoop/
capacity-scheduler.xml      100% 9213      1.2MB/s   00:00
configuration.xml           100% 1335      216.5KB/s 00:00
container-executor.cfg      100% 2567      388.4KB/s 00:00
core-site.xml               100% 870       123.1KB/s 00:00
hadoop-env.cmd              100% 3999      420.3KB/s 00:00
hadoop-env.sh               100% 16KB       9.8MB/s   00:00
hadoop-metrics2.properties  100% 3321      4.2MB/s   00:00
hadoop-policy.xml           100% 14KB      12.8MB/s   00:00
hadoop-user-functions.sh.example 100% 3414      454.3KB/s 00:00
hdfs-rbf-site.xml           100% 683       473.0KB/s 00:00
hdfs-site.xml               100% 1071      954.4KB/s 00:00
httpfs-env.sh               100% 1484      82.0KB/s   00:00
httpfs-log4j.properties    100% 1657      2.5MB/s   00:00
httpfs-site.xml             100% 620       1.0MB/s   00:00
kms-acls.xml                100% 3518      354.5KB/s 00:00
kms-env.sh                  100% 1351      2.0MB/s   00:00
kms-log4j.properties        100% 1860      2.5MB/s   00:00
kms-site.xml                100% 682       956.8KB/s 00:00
log4j.properties            100% 14KB      16.1MB/s   00:00
mapred-env.cmd              100% 951       1.5MB/s   00:00
mapred-env.sh               100% 1764      3.3MB/s   00:00
mapred-queues.xml.template  100% 4113      5.5MB/s   00:00
mapred-site.xml             100% 758       941.5KB/s 00:00
```

```
hadoopuser@hadoop-master: /home/hadoop
hadoopuser@hadoop-master: /home/hadoop$ scp /usr/local/hadoop/etc/hadoop/* hadoop
-slave3:/usr/local/hadoop/etc/hadoop/
capacity-scheduler.xml      100% 9213      3.8MB/s   00:00
configuration.xml           100% 1335      960.2KB/s 00:00
container-executor.cfg      100% 2567      2.5MB/s   00:00
core-site.xml               100% 870       953.4KB/s 00:00
hadoop-env.cmd              100% 3999      4.7MB/s   00:00
hadoop-env.sh               100% 16KB       8.7MB/s   00:00
hadoop-metrics2.properties  100% 3321      5.2MB/s   00:00
hadoop-policy.xml           100% 14KB      18.8MB/s   00:00
hadoop-user-functions.sh.example 100% 3414      3.7MB/s   00:00
hdfs-rbf-site.xml           100% 683       1.4MB/s   00:00
hdfs-site.xml               100% 1071      2.2MB/s   00:00
httpfs-env.sh               100% 1484      2.1MB/s   00:00
httpfs-log4j.properties    100% 1657      2.9MB/s   00:00
httpfs-site.xml             100% 620       1.2MB/s   00:00
kms-acls.xml                100% 3518      4.6MB/s   00:00
kms-env.sh                  100% 1351      1.8MB/s   00:00
kms-log4j.properties        100% 1860      3.2MB/s   00:00
kms-site.xml                100% 682       1.2MB/s   00:00
log4j.properties            100% 14KB      14.8MB/s   00:00
mapred-env.cmd              100% 951       1.5MB/s   00:00
mapred-env.sh               100% 1764      2.6MB/s   00:00
mapred-queues.xml.template  100% 4113      2.0MB/s   00:00
mapred-site.xml             100% 758       1.3MB/s   00:00
```

28. Formateamos el sistema de archivos de HDFS.

```
hadoopuser@hadoop-master:~$ source /etc/environment
hadoopuser@hadoop-master:~$ hdfs namenode -format
WARNING: /usr/local/hadoop/logs does not exist. Creating.
2025-11-06 19:58:46,215 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = hadoop-master/172.21.173.135
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.4.2
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/
hadoop/common/lib/jetty-server-9.4.57.v20241219.jar:/usr/local/hadoop/share/hado
op/common/lib/netty-resolver-dns-native-macos-4.1.118.Final-osx-aarch_64.jar:/us
r/local/hadoop/share/hadoop/common/lib/netty-buffer-4.1.118.Final.jar:/usr/local
/hadoop/share/hadoop/common/lib/jsp-api-2.1.jar:/usr/local/hadoop/share/hadoop/c
ommon/lib/commons-io-2.16.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-
servlet-9.4.57.v20241219.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-tra
nsport-classes-kqueue-4.1.118.Final.jar:/usr/local/hadoop/share/hadoop/common/li
b/commons-daemon-1.0.13.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-tran
sport-native-epoll-4.1.118.Final-linux-riscv64.jar:/usr/local/hadoop/share/hadoo
p/common/lib/kerby-asn1-2.0.3.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb
-crypto-2.0.3.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-xml-9.4.57.v20
241219.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-http-9.4.57.v20241219
```

29. Encendemos HDFS.

```
hadoopuser@hadoop-master:~$ start-dfs.sh
Starting namenodes on [hadoop-master]
Starting datanodes
hadoop-slave3: WARNING: /usr/local/hadoop/logs does not exist. Creating.
hadoop-slave1: WARNING: /usr/local/hadoop/logs does not exist. Creating.
hadoop-slave2: WARNING: /usr/local/hadoop/logs does not exist. Creating.
Starting secondary namenodes [hadoop-master]
```

Comprobamos si se ha encendido correctamente.

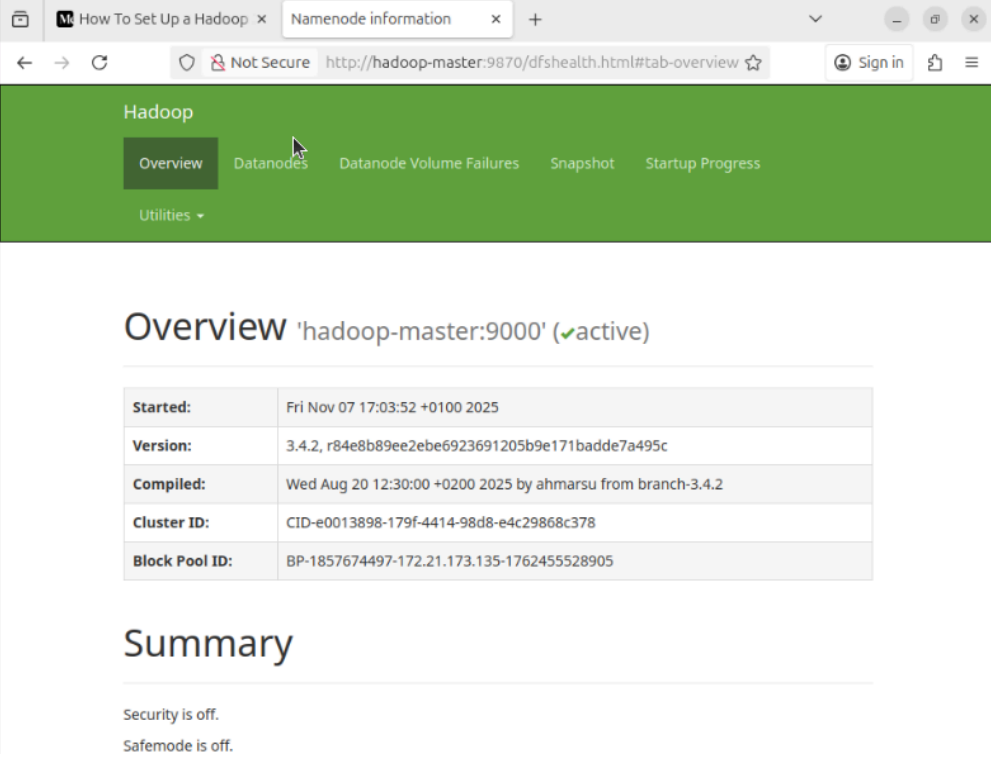
```
hadoopuser@hadoop-master:~$ jps
5907 SecondaryNameNode
6039 Jps
5655 NameNode
```

```
hadoop@hadoop-slave1:~$ jps
5420 Jps
```

```
hadoop@hadoop-slave2:~$ jps
5903 Jps
```

```
hadoop@hadoop-slave3:~$ jps
5473 Jps
```

30. Ponemos en el navegador hadoop-master:9870



The screenshot shows a web browser window with the URL `http://hadoop-master:9870/dfshealth.html#tab-overview`. The page title is "Hadoop" and the main navigation bar includes "Overview", "Datanodes", "Datanode Volume Failures", "Snapshot", and "Startup Progress". The "Overview" tab is selected. Below the navigation bar, the page displays "Overview 'hadoop-master:9000' (✓active)". A table provides details about the NameNode:

Started:	Fri Nov 07 17:03:52 +0100 2025
Version:	3.4.2, r84e8b89ee2ebe6923691205b9e171badde7a495c
Compiled:	Wed Aug 20 12:30:00 +0200 2025 by ahmarsu from branch-3.4.2
Cluster ID:	CID-e0013898-179f-4414-98d8-e4c29868c378
Block Pool ID:	BP-1857674497-172.21.173.135-1762455528905

Below the table, a "Summary" section indicates that "Security is off." and "Safemode is off."

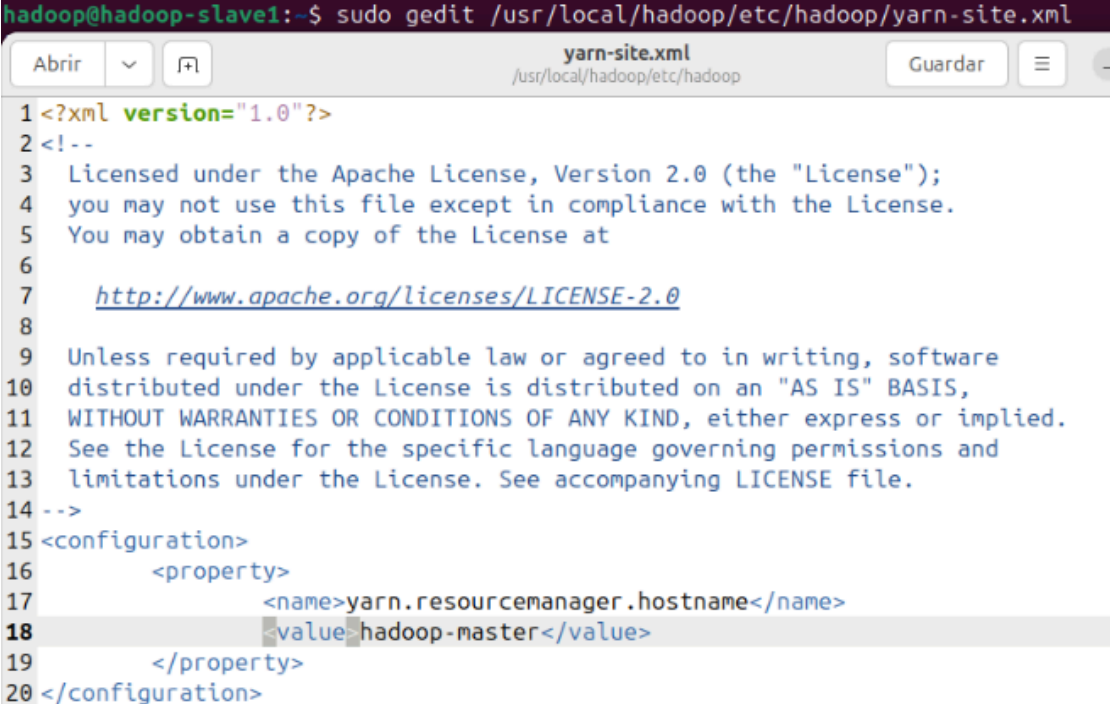
31. Configuramos yarn con los siguientes comandos:

```
export HADOOP_HOME="/usr/local/hadoop"
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export HADOOP_HDFS_HOME=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_YARN_HOME=$HADOOP_HOME
```

```
hadoopuser@hadoop-master:~$ export HADOOP_HOME="/usr/local/hadoop"
hadoopuser@hadoop-master:~$ export HADOOP_COMMON_HOME=$HADOOP_HOME
hadoopuser@hadoop-master:~$ export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
hadoopuser@hadoop-master:~$ export HADOOP_HDFS_HOME=$HADOOP_HOME
hadoopuser@hadoop-master:~$ export HADOOP_MAPRED_HOME=$HADOOP_HOME
hadoopuser@hadoop-master:~$ export HADOOP_YARN_HOME=$HADOOP_HOME
```

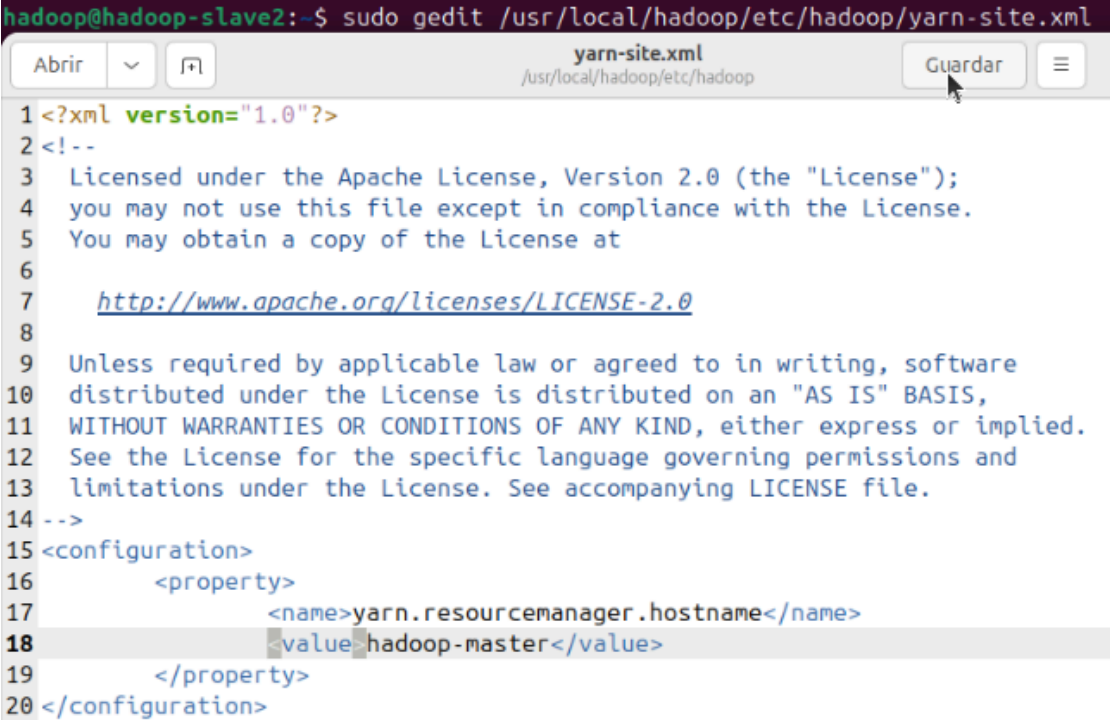

32. En los slaves abrimos el archivo yarn-site.xml y añadimos lo siguiente:

```
<property>
  <name>yarn.resourcemanager.hostname</name>
  <value>hadoop-master</value>
</property>
```



The screenshot shows a terminal window on a Hadoop slave node (hadoop@hadoop-slave1) where the file /usr/local/hadoop/etc/hadoop/yarn-site.xml is being edited with gedit. The file content is as follows:

```
1 <?xml version="1.0"?>
2 <!--
3   Licensed under the Apache License, Version 2.0 (the "License");
4   you may not use this file except in compliance with the License.
5   You may obtain a copy of the License at
6
7   http://www.apache.org/licenses/LICENSE-2.0
8
9   Unless required by applicable law or agreed to in writing, software
10  distributed under the License is distributed on an "AS IS" BASIS,
11  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
12  See the License for the specific language governing permissions and
13  limitations under the License. See accompanying LICENSE file.
14 -->
15 <configuration>
16   <property>
17     <name>yarn.resourcemanager.hostname</name>
18     <value>hadoop-master</value>
19   </property>
20 </configuration>
```

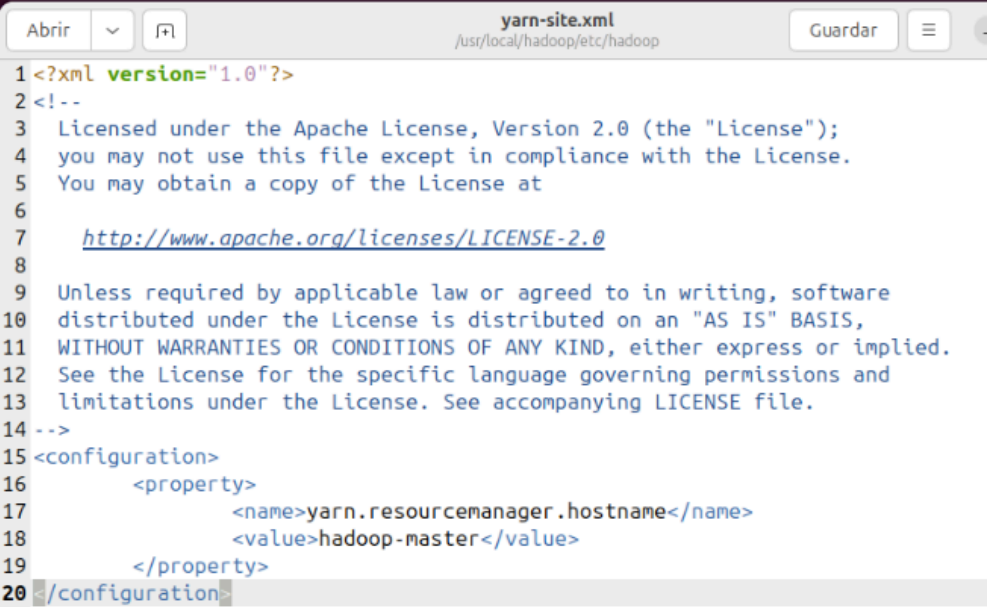


The screenshot shows a terminal window on a Hadoop slave node (hadoop@hadoop-slave2) where the file /usr/local/hadoop/etc/hadoop/yarn-site.xml is being edited with gedit. The file content is as follows:

```
1 <?xml version="1.0"?>
2 <!--
3   Licensed under the Apache License, Version 2.0 (the "License");
4   you may not use this file except in compliance with the License.
5   You may obtain a copy of the License at
6
7   http://www.apache.org/licenses/LICENSE-2.0
8
9   Unless required by applicable law or agreed to in writing, software
10  distributed under the License is distributed on an "AS IS" BASIS,
11  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
12  See the License for the specific language governing permissions and
13  limitations under the License. See accompanying LICENSE file.
14 -->
15 <configuration>
16   <property>
17     <name>yarn.resourcemanager.hostname</name>
18     <value>hadoop-master</value>
19   </property>
20 </configuration>
```



```
hadoop@hadoop-slave3:~$ sudo gedit /usr/local/hadoop/etc/hadoop/yarn-site.xml
```



```

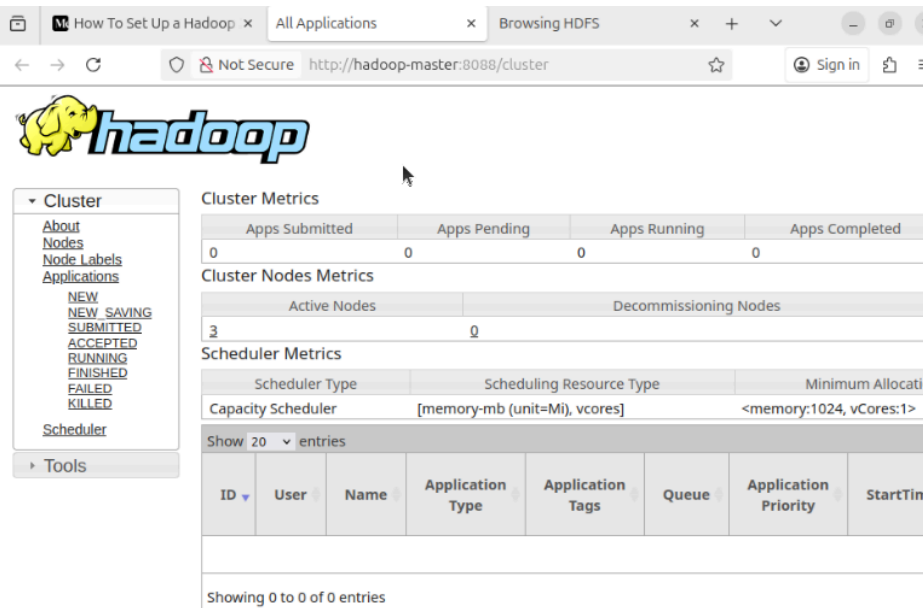
1 <?xml version="1.0"?>
2 <!--
3 Licensed under the Apache License, Version 2.0 (the "License");
4 you may not use this file except in compliance with the License.
5 You may obtain a copy of the License at
6
7 http://www.apache.org/licenses/LICENSE-2.0
8
9 Unless required by applicable law or agreed to in writing, software
10 distributed under the License is distributed on an "AS IS" BASIS,
11 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
12 See the License for the specific language governing permissions and
13 limitations under the License. See accompanying LICENSE file.
14 -->
15 <configuration>
16   <property>
17     <name>yarn.resourcemanager.hostname</name>
18     <value>hadoop-master</value>
19   </property>
20 </configuration>

```

33. Encendemos yarn.

```
hadoopuser@hadoop-master:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

34. Escribimos en el navegador: <http://hadoop-master:8088/cluster>



Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed
0	0	0	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes
3	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation
Capacity Scheduler	[memory-mb (unit=Mb), vcores]	<memory:1024, vCores:1>

Show 20 entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime
Showing 0 to 0 of 0 entries							

Conclusiones

Mis conclusiones son que instalar hadoop es un trabajo con muchos pasos que hay que seguir al pie de la letra para que no haya ningún error. Hay que tener claro cuáles son las IPs de cada máquina y separarlas bien. Por lo demás, mientras se sigan bien los pasos, no es una instalación complicada.

Problemas encontrados/soluciones

En mi caso no he tenido ningún problema, he seguido cada paso de la instalación y ha funcionado todo correctamente.